

62. Ans: (a)
Sol: Water

$$\rho = 1000 \text{ kg/m}^3, \\ P = 200 \text{ bar}, \\ P.T = 0.05 \text{ sec},$$

Liquid metal
 $\rho = 2000 \text{ kg/m}^3,$
 $P = 400 \text{ bar}$
 $P.T = ?$

$$P.T = \frac{\text{Volume}}{\text{Area} \times \text{Velocity}} = \frac{\text{Volume}}{A \times \sqrt{2gh}}$$

$$\frac{(P.T)_{\text{water}}}{(P.T)_{\text{liquid metal}}} = \frac{\left(\frac{\text{Volume}}{A \times \sqrt{2gh}}\right)_{\text{water}}}{\left(\frac{\text{Volume}}{A \times \sqrt{2gh}}\right)_{\text{liquid metal}}} = 1$$

$$\frac{(P.T)_{\text{water}}}{(P.T)_{\text{liquid metal}}} = 1 \Rightarrow \frac{0.05}{(P.T)_{\text{liquid metal}}} = 1$$

$$\Rightarrow (P.T)_{\text{liquid metal}} = 0.05 \text{ sec}$$

63. Ans: (a)
Sol: Blow hole

- $\Rightarrow$ Poor Permeability
- $\Rightarrow$ Insufficient Fluidity
- Hot tearing $\Rightarrow$ Poor Collapsibility
- Wash $\Rightarrow$ Mold erosion

64. Ans: 28.92 (Range: 28.80 to 29.00)

Sol: $\frac{\text{Volume}}{t} = A_c \sqrt{2gh_c}$

At point (3) is open to atmosphere so (3) is the choke.

$$g = 9.81 \text{ m/sec}^2 = 981 \text{ cm/sec}^2$$

$$\frac{40 \times 30 \times 15}{t} = \frac{\pi}{4} \times 2^2 \sqrt{2 \times 981 \times 20}$$

$$\Rightarrow \frac{18000}{t} = 622.32088$$

$$\Rightarrow t = 28.923 \text{ sec}$$

65. Ans: 29.3 (Range: 28 to 31)

Sol: $k = 2 \text{ sec/mm}^2; \tau = k \left(\frac{V}{A}\right)^2; A \propto \frac{1}{\sqrt{\tau}}$

$$\frac{A_2}{A_1} = \sqrt{\frac{\tau_1}{\tau_2}} = \frac{1}{\sqrt{2}} = 0.707$$

$$\Rightarrow (A_2 - A_1) = (1 - 0.707) = 0.293 = 29.3\%$$

66. Ans: (b)

Sol: $N^2 D = \text{Constant}$

$$D = \frac{0.275 + 0.250}{2} = 0.2625 \text{ m}$$

$$N^2 D = \frac{a}{2\pi^2}$$

$$N^2 D = \frac{65g}{2\pi^2}$$

$$N^2 \times 0.2625 = \frac{65 \times 9.81}{2\pi^2}$$

$$N = 11.09 \text{ r.p.s}$$

$$N = 665.599 \text{ r.p.m} \approx 650 \text{ r.p.m}$$

67. Ans: 4.28

Sol: Given, Volume = 1200 cm³

$$h_c = 10 \text{ cm};$$

$$A_c = 2 \text{ cm}^2; g = 9.81 \text{ m/sec}^2$$

$$t = ?$$

$$\frac{\text{volume}}{t} = A_c \sqrt{2gh_c}$$

$$\frac{1200}{t} = 2\sqrt{2 \times 9.81 \times 10}$$

$\Rightarrow$ The time taken to fill the mould cavity (t) = 4.284 sec

68. Ans: 36.6

Sol: τ = solidification time

$$\tau_r = 1.25 \times M_c^2$$

$$M_r^2 = 1.25 \times M_c^2$$

$$M_r = \sqrt{1.25} \times M_c = 1.118 M_c$$

$$\frac{D}{6} = 1.118 M_c$$

$$D = 6 \times 1.118 \times M_c$$

where,

$$M_c = \frac{\text{Vol}}{A_s} = \frac{60 \times 40 \times 20}{2[(60 \times 40) + (40 \times 20) + (20 \times 60)]}$$

$$M_c = \frac{48000}{8800} = 5.4545$$

$$\text{Now, } D = 6 \times 1.118 \times 5.4545 = 36.6 \text{ cm}$$

69. Ans: 662 (Range 660 to 664)

Sol: Given, True centrifugal casting,

Outer diameter, $D_o = 250$ mm

Inner diameter, $D_i = 230$ mm

Acceleration due to gravity (g) = 10 m/s²

G factor (G) = 60, $N = ?$

$$G = \frac{mR_m \omega^2}{mg}$$

$$\Rightarrow 60 = \frac{D_o}{2g} \left(\frac{2\pi N}{60} \right)^2$$

$$\Rightarrow 60 = \frac{0.25}{2 \times 10} \left(\frac{2\pi N}{60} \right)^2$$

Required Rotational Speed (N) = 662 rpm

70. Ans: 2.82 (Range 2.60 to 3.00)

Sol: Given,

Case 1: Casting 1 product volume

$$(V_1) = 75 \times 125 \times 20 \text{ mm}^3$$

Solidification time (τ_1) = 2.0 min

Index, $n = 2$

Case 2: Casting 2 product volume

$$(V_2) = \frac{\pi}{4} D^2 \times h$$

$$(V_2) = \frac{\pi}{4} \times 50^2 \times 50$$

$$(\tau_2) = ?$$

We know, $\frac{\tau_1}{\tau_2} = \left(\frac{M_1}{M_2} \right)^2$

$$M_1 = \frac{V_1}{A_1} = \frac{75 \times 125 \times 20}{2[(75 \times 125) + (125 \times 20) + (200 \times 75)]}$$

$$= 7.0093$$

$$M_2 = \frac{V_2}{A_2} = \frac{D}{6} = \frac{50}{6} = 8.333$$

$$\frac{\tau_1}{\tau_2} = \left(\frac{M_1}{M_2} \right)^2 \Rightarrow \frac{2.0}{\tau_2} = \left(\frac{7.0093}{8.333} \right)^2$$

$$\tau_2 = 2.377 \text{ min}$$

Solidification time cylinder, $\tau_2 \approx 2.38$ min

71. Ans: 75 (Range 74.5 to 75.5)

Sol: $t_1 = B(M_1)^2 \rightarrow M_1 = \frac{d_1}{6} = \frac{1}{6} = 0.16 \text{ m}$

$$t_2 = B(M_2)^2 \rightarrow M_2 = \frac{d_2}{6} = \frac{0.5}{6} = 0.083 \text{ m}$$

$$\frac{t_1}{t_2} = \frac{M_1^2}{M_2^2} \Rightarrow \frac{t_1}{t_2} = \frac{0.16^2}{0.08^2} = 4$$

$$t_1 = 4 t_2 \Rightarrow t_2 = 0.25 t_1$$

Percentage reduction in solidification time

$$= \frac{t_1 - t_2}{t_1}$$

$$= \frac{t_1 - 0.25 t_1}{t_1} = 0.75$$

Percentage reduction = 75 %

72. Ans: 93.21

Sol: Volume of cubical casting = 100^3

Volume after solidification shrinkage = $100^3 \times 0.9$

Volume after solid contraction = $100^3 \times 0.9 \times 0.9$

$$= 8,10,000$$

$$\therefore a^3 = 810000 \Rightarrow a = 93.21 \text{ mm}$$

Side after cubical cost = 93.21 mm

73. Ans: 3

Sol: $d = 10$ cm, $\rho = 7.85 \times 10^{-3}$ kg/cm³

$m = 12.56$ kg, $n = 2$, $\tau = 12$ min, $K_1 = ?$

L - Length of casting

$$m = \rho \times \text{vol}$$

$$\text{vol} = \frac{m}{\rho}$$

$$\frac{\pi}{4} d^2 \times L = \text{Vol}$$

$$\frac{\pi}{4} \times 10^2 \times L = \frac{12.56}{7.85 \times 10^{-3}}$$

$$L = 20.37 \text{ cm}$$

$$C = m^2 K_1 \rightarrow T_o = \left(\frac{V}{A_s} \right)^2 K_1$$

$$12 = \left(\frac{\frac{\pi}{4} \times 10^2 \times 2037}{\pi \times 10 \times 20.37 + 2 \times \frac{\pi}{2} \times 1^2} \right)^2 K_1$$

$$12 = \left(\frac{1600}{979} \right)^2 K_1$$

$$K_1 = 3 \text{ min/cm}^2$$

One Mark Questions

10. Penetration is increased by (GATE-ME-90)
 (a) Increasing welding current and welding speed
 (b) Increasing welding current and decreasing welding speed
 (c) Decreasing welding current and welding speed
 (d) Decreasing welding current and increasing welding speed
11. At small variations of arc length at operating conditions, the manual metal arc welding transformer provides nearly (GATE-ME-90)
 (a) Constant power (b) Constant power factor
 (c) Constant voltage (d) Constant current
12. For gas welding a particular job using a *neutral oxy-acetylene flame* the acetylene consumption was 10 litres. The oxygen consumption from the cylinder in liters will be (GATE-ME-91)
 (a) 5 (b) 10 (c) 15 (d) 20
13. For resistance spot welding of 1.5 mm thick steel sheets, the current required is of the order (GATE-ME-92)
 (a) 10 Amp (b) 100 Amp
 (c) 1000 Amp (d) 10,000 Amp
14. In an explosive welding process, the (maximum/minimum) velocity of impact is fixed by the velocity of sound in the (flyer/target) plate material. (GATE-ME-92)
15. In DC welding, the straight polarity (electrode negative) results in (GATE-ME-93)
 (a) Lower penetration
 (b) Lower deposition rate
 (c) Less heating of work piece
 (d) Smaller weld pool
16. The electrodes used in arc welding are coated. This is not expected to (GATE-ME-94)
 (a) Provide protective atmosphere to weld
 (b) Stabilize the arc
 (c) Add alloying elements
 (d) Prevent electrode from contamination
17. The ratio of acetylene to oxygen is approximately for *neutral flame* used in gas welding. (GATE-ME-94)
18. Generally cylindrical parts produced by powder metallurgy should not have non-uniform cross section and length to diameter ratio exceeding (GATE-ME-94)
19. Preheating before welding is done to (GATE-ME-96)
 (a) Make the steel softer
 (b) Burn away oil, grease etc from the plate surfaces
 (c) Prevent cold cracks
 (d) Prevent plate distortion
20. Two plates of the same metal having equal thickness are to be butt welded with electric arc. When the plate thickness changes, welding is achieved by (GATE-ME-01)
 (a) Adjusting the current
 (b) Adjusting the duration of the current
 (c) Changing the electrode size
 (d) Changing the electrode coating
21. The temperature of a carburizing flame in gas welding is..... that of a neutral or an oxidizing flame. (GATE-ME-02)
 (a) Lower than (b) Higher than
 (c) Equal to (d) Unrelated to
22. Which of the following arc welding processes does not use consumable electrode? (GATE-ME-02)
 (a) GMAW (b) GTAW (c) SAW (d) None

14. In oxyacetylene gas welding, temperature at the inner cone of the flame is around **(GATE-ME-03)**
 (a) 3500°C (b) 3200°C
 (c) 2900°C (d) 2550°C
15. The strength of a brazed joint **(GATE-ME-05)**
 (a) Decreases with increase in gap between the two joining surfaces
 (b) Increases with increase in gap between the two joining surfaces
 (c) Decreases up to certain gap between the two joining surfaces beyond which it increases
 (d) Increases up to certain gap between the two joining surfaces beyond which it decreases
16. Which of the following is a solid state welding process? **(GATE-ME-07)**
 (a) GTAW
 (b) Resistance spot welding
 (c) Friction welding
 (d) SAW
17. Which one among the following welding processes uses non-consumable electrode? **(GATE-ME-11)**
 (a) Gas metal arc welding
 (b) Submerged arc welding
 (c) Gas tungsten arc welding
 (d) Flux coated arc welding
18. The major difficulty during welding of aluminium is due to its **(GATE-ME-14-SET-1)**
 (a) high tendency of oxidation
 (b) high thermal conductivity
 (c) low melting point
 (d) low density
19. In solid-state welding, the contamination layers between the surfaces to be welded are removed by **(GATE-ME-14-SET-1)**
 (a) alcohol (b) plastic deformation
 (c) water jet (d) sand blasting
20. Within the Heat Affected Zone (HAZ) in a fusion welding process, the work material undergoes **(GATE-ME-14-SET-4)**
 (a) microstructural changes but does not melt
 (b) neither melting nor microstructural changes
 (c) both melting and microstructural changes after solidification
 (d) melting and retains the original microstructure after solidification
21. In welding brass with oxy-acetylene flame, the type of flame used is **(GATE-PI-89)**
22. The mode of metal transfer in CO₂-MIG welding at low temperature is **(GATE-PI-90)**
 (a) Spray transfer
 (b) Short circuiting type of transfer
 (c) Globular transfer
 (d) Droplet transfer
23. With increasing joint thickness, the tensile strength of a brazed joint **(GATE-PI-90)**
 (a) Continuously decreases
 (b) First decreases and then increases
 (c) Continuously increases
 (d) First increases and then decreases
24. Match the following lists:
List - I (Welding Process)
 A. Thermit welding
 B. Projection welding
 C. MIG welding
 D. Friction welding
List - II (Heat Source)
 1. Electric arc
 2. Mechanical work
 3. Exothermic chemical reaction
 4. Ohmic resistance **(GATE-PI-90)**
25. In TIG welding aA. ... andB.... electrode is used. **(GATE-PI-90)**
 A. non-consumable / consumable
 B. coated / bare

26. High alloy steel components are preheated before welding for reducing (GATE-PI-91)
 (a) Heat affected zone
 (b) Total energy consumption
 (c) Total time of welding
 (d) Welding stresses

27. In spot welding of aluminum sheets, the welding current should be.....A.... and weld time should be keptB..... than for MS sheet of same thickness. (GATE-PI-91)
 A. higher / lower B. higher / lower

28. Continuous rails for Indian railways are welded by Welding process. (GATE-PI-91)

29. is the non-fusion method of joining dissimilar metals with metals whose melting temperature is greater than 400°C. (GATE-PI-91)

30. The joint configuration best suited for adhesive bonding is (GATE-PI-92)
 (a) butt (b) fillet (c) lap (d) spot

31. The type of coated electrode most widely used for welding low carbon steel is (GATE-PI-92)
 (a) cellulose (b) acidic (c) rutile (d) oxide

32. Filler material is A... in resistance welding and the heat generated in the process is directly proportional toB..... (GATE-PI-92)
 A. used / not used
 B. square of the current / cube of the current

33. The composition of an oxy-acetylene flame for cutting of steel contain more oxygen (T/F). (GATE-PI-94)

34. The consumption of an oxy-acetylene flame for cutting of steel contains more oxygen (T/F). (GATE-PI-95)

35. Which of the following powders should be fed for effective oxy-fuel cutting of stainless steel? (GATE-PI-01)
 (a) Steel (b) Aluminum (c) Copper (d) Ceramic

36. High speed electron beam welding is focused on the weld spot using (GATE-PI-03)
 (a) vacuum lens (b) inert gas lens
 (c) optical lens (d) magnetic lens

37. In resistance welding, heat is generated due to the resistance between (GATE-PI-03)
 (a) electrode and work piece
 (b) asperities between touching plates
 (c) two dissimilar metals being in contact
 (d) inter-atomic forces

38. The current in Amperes used in resistance spot welding of plain carbon steel sheets (1 to 3 mm thick) lies within the range: (GATE-PI-04)
 (a) 10-50 (b) 50-500 (c) 500-5000 (d) 5000-50000

39. The metal powder used in Thermit welding of steel is (GATE-PI-04)
 (a) Al (b) Cu (c) Pb (d) W

40. Which of the following welding process results in the smallest heat affected zone? (GATE-PI-11)
 (a) Shielded metal arc welding
 (b) Gas welding
 (c) Laser beam welding
 (d) Thermit welding

41. In resistance seam welding, the electrode is in the form of a (GATE-PI-11)
 (a) cylinder (b) flat plate
 (c) coil of wire (d) circular disc

42. Brazing and soldering are (GATE-PI-14)
 (a) plastic joining methods
 (b) homogeneous joining methods
 (c) autogenous joining methods
 (d) heterogeneous joining methods

43. In a linear arc welding process, the heat input per unit length is inversely proportional to
(a) welding current (GATE-15-Set-1)
(b) welding voltage
(c) welding speed
(d) duty cycle of the power source
44. Which two of the following joining processes are autogeneous?
(i) Diffusion welding
(ii) Electro slag welding
(iii) Tungsten inert gas welding
(iv) Friction welding (GATE-15-Set 3)
(a) (i) and (iv) (b) (ii) and (iii)
(c) (ii) and (iv) (d) (i) and (iii)
45. Under optimal conditions of the process the temperatures experienced by a copper work piece in fusion welding, brazing and soldering are such that (GATE - 16 - SET - 1)
(a) $T_{\text{welding}} > T_{\text{soldering}} > T_{\text{brazing}}$
(b) $T_{\text{soldering}} > T_{\text{welding}} > T_{\text{brazing}}$
(c) $T_{\text{brazing}} > T_{\text{welding}} > T_{\text{soldering}}$
(d) $T_{\text{welding}} > T_{\text{brazing}} > T_{\text{soldering}}$
46. In an arc welding process, welding speed is doubled. Assuming all other process parameters to be constant, the cross sectional area of the weld bead will (GATE - 17 - SET - 1)
(a) increase by 20 % (b) increase by 50 %
(c) reduce by 25 % (d) reduce by 50 %
47. In gas tungsten arc welding process, the material coated on pure tungsten electrode to enhance its current carrying capacity is (GATE - PI - 17)
(a) Titanium (b) Manganese (c) Radium (d) Thorium
48. In an oxy-acetylene gas welding process, oxygen and acetylene are mixed in a ratio of 1.5:1 (by volume). The flame is (PI_GATE - 18)
(a) neutral (b) carburizing
(c) reducing (d) oxidizing
49. Which one of the following welding methods provides the highest heat flux (W/mm^2)? (GATE-19-SET-1)
(a) Tungsten inert gas welding
(b) Laser beam welding
(c) Plasma arc welding
(d) Oxy-acetylene gas welding
50. The correct statement pertaining to the friction welding process is (GATE - 19_PI)
(a) Heat affected zone is not formed
(b) Flashes are not produced
(c) Dissimilar materials cannot be joined
(d) Melting of the base material(s) is not involved
51. Two plates, each of 6 mm thickness, are to be butt-welded. Consider the following processes and select the correct sequence in increasing order of size of the heat affected zone.
1. Arc welding
2. MIG welding
3. Laser beam welding
4. Submerged arc welding (GATE-20-SET-2)
(a) 1-4-2-3 (b) 3-4-2-1 (c) 3-2-4-1 (d) 4-3-2-1
52. Which one of the following defects is NOT associated with welding processes? (GATE_PI-21)
(a) Hydrogen embrittlement (b) Angular distortion
(c) Earring (d) Hot tear
53. Two surfaces P and Q are to be joined together. In which of the given joining operation(s), there is no melting of the two surfaces P and Q for creating the joint? (GATE_ME-23)
(a) Arc welding (b) Brazing
(c) Adhesive bonding (d) Spot welding
54. Two metal sheets are joined using resistance spot welding. A welding current of 4500 A is applied for 0.2 s. The effective contact resistance at the sheet interface is $400 \times 10^{-6} \Omega$. The thermal efficiency of the welding process is 50 %. The amount of heat, in J, used for producing a spot weld is _____ (in integer). (GATE_PI-23)

Two Marks Questions

Q1. Match the following:

List - I (Welding process)

- | | |
|--------------------|---------------------|
| A. Thermit welding | B. Seam welding |
| C. MIG welding | D. Friction welding |

List - II (Heat source)

1. Chemical reaction
2. Exothermic chemical reaction
3. Electric spark
4. Ohmic resistance
5. Mechanical work
6. Electric arc

(GATE-ME-91)

Q2. Match the following:

List - I

- A. Welding of aluminum alloy
- B. Ship building
- C. Joining of HSS drill bit to shank
- D. Deep penetration precision welds

List - II

1. Submerged arc welding
2. Electron beam welding
3. TIG welding
4. Friction welding
5. Gas welding

(GATE-ME -96)

Q3. Resistance spot welding is performed on two plates of 1.5 mm thickness with 6 mm diameter electrode, using 15000 A current for a time duration of 0.25 sec. Assuming the interface resistance to be 0.0001Ω . The heat generated to form the weld is

(GATE-ME-01)

- | | |
|-----------------|-----------------|
| (a) 5625 W-sec | (b) 8437 W-sec |
| (c) 22500 W-sec | (d) 33750 W-sec |

Q4. Match the following:

List - I (Work material)

- P. Aluminum
- Q. Die steel
- R. Copper wire
- S. Titanium sheet

List - II (Type of joining)

1. Submerged Arc Welding
2. Soldering
3. Thermit Welding
4. Atomic Hydrogen Welding
5. Gas Tungsten-Arc welding
6. Laser Beam Welding

	P	Q	R	S
(a)	2	5	1	3
(b)	6	3	4	2
(c)	4	1	6	2
(d)	5	4	2	6

Q5. Two 1 mm thick steel sheets are to be spot welded at a current of 5000 A. Assuming effective resistance to be 200 micro-ohms and current flow time of 0.2 second, heat generated during the process will be

(GATE-ME -04)

- | | |
|---------------|-----------------|
| (a) 0.2 Joule | (b) 1 Joule |
| (c) 5 Joule | (d) 1000 Joules |

Q6. Spot welding of two 1 mm thick sheets of steel (density = 8000 kg/m^3) is carried out successfully by passing a certain amount of current for 0.1 second through the electrodes. The resultant weld nugget formed is 5 mm in diameter and 1.5 mm thick. If the latent heat of fusion of steel is 1400 kJ/kg and the effective resistance in the welding operation is $200 \mu\Omega$, the current passing through the electrodes is approximately

(GATE-ME-05)

- (a) 1480 A (b) 3300 A (c) 4060 A (d) 9400 A

Q7. In arc welding process, the voltage and current are 25 V and 300 A respectively. The arc heat transfer efficiency is 0.85 and welding speed is 8 mm/sec. The net heat input (in J/mm) is

(GATE-ME-06)

- (a) 64 (b) 797 (c) 1103 (d) 79700

Q8. A DC welding machine with a linear power source characteristic provides open circuit voltage of 80 V and short circuit current of 800 A. During welding with the machine, the measured arc current is 500

A corresponding to an arc length of 5.0 mm and the measured arc current 460 A corresponding to an arc length of 7.0 mm. The linear voltage (E) & arc length (L) characteristic of the welding arc can be given as (where E is in Volts and L is in mm)

(GATE-ME-07)

- (a) $E = 20 + 2L$ (b) $E = 20 + 8L$
(c) $E = 80 + 2L$ (d) $E = 80 + 8L$

09. Two metallic sheets, each of 2.0 mm thickness are welded in lap joint configuration by resistance spot welding at a welding current of 10 kA and welding time of 10 milliseconds. A spherical fusion zone extending up to the full thickness of each sheet is formed. The properties of the metallic sheets are given below:

Ambient temperature = 293 K

Melting point temperature = 1793 K

Density = 7000 kg/m³

Latent heat of fusion = 300 kJ/kg

Specific heat = 800 J/kgK

Assume:

- Contact resistance along sheet – sheet interface is 500 micro-ohm and along electrode –sheet interface is zero.;
- No conductive heat loss through the bulk sheet material; and
- The complete weld fusion zone is at the melting temperature.

The melting efficiency (in %) of the process is.....

(GATE-ME-07)

- (a) 50.37 (b) 60.37 (c) 70.37 (d) 80.37

10. In arc welding of a butt joint, the welding speed is to be selected such that highest cooling rate is achieved. Melting efficiency and heat transfer efficiency are 0.5 and 0.7, respectively. The area of the weld cross section is 5 mm² and the unit energy required to melt the metal is 10 J/mm³. If the welding power is 2 kW, the welding speed in mm/s is closet to

(GATE-ME-08)

- (a) 4 (b) 14 (c) 24 (d) 34

11. Two steel bars each of diameter 10 mm are coaxially friction welded end to end at an axial pressure of 200 MPa and rotational speed of 4000 rpm. The coefficient of friction between the mating faces of the rotating bars is 0.50. The torque is assumed to act at the 3/4th radius of the rotating bar. The power (in kW) consumed at the interface of the welding is

(ME-GATE-10)

- (a) 12.33 (b) 16.44 (c) 18.50 (d) 24.66

12. In welding a mild steel sheet using an oxy-acetylene flame, the total amount of acetylene consumed was 8.0 ltrs. The oxygen consumption from the cylinder is

(GATE-PI-89)

- (a) 8.0 ltrs (b) 16 ltrs (c) 20 ltrs (d) 24 ltrs

13. Match the following:

(GATE-PI-89)

List -I

List -II

A. Cellulose coating

1. Alloying agent

B. Rutile coating

2. Absorb gasses

C. Iron powder coating

3. Forming slag

D. Basic coating

4. Shielding gasses

14. In welding process, penetration is increased by

(GATE-PI-91)

- (a) increasing current and decreasing speed
(b) decreasing both current and speed
(c) increasing both arc voltage and speed
(d) increasing arc voltage and decreasing current

15. Match the following:

List - I (Application)

A. Highly reactive metals

B. Long straight butt joint in thick sections

C. Site weld of rails

D. MS rods to plate like components

List - II (Welding Process)

1. Ultrasonic

2. Thermit

3. Electron beam

4. Submerged arc

5. Stud arc

(GATE-PI-92)

16. During gas welding of copper sheets a neutral flame was used consuming 8 liters of acetylene. But the weld was found to be defective. So it was re-welded using an oxidizing flame. The likely consumption of oxygen in liters will be (GATE-PI-93)

(a) 6 (b) 8 (c) 10 (d) 16

17. In spot pulsed laser welding of aluminum plates (density = 2700 kg/m^3 , specific heat = 896 J/kg , melting temp = 933 K , latent heat of melting = 398 kJ/kg) at a temperature of 30°C , pulse with energy of 0.5 J is focused onto an area of 0.05 mm^2 . If the entire energy is coupled into the material, what will be the depth of weld assuming the CS area of the weld is circular and is uniform throughout its depth and only heat conduction in the direction of penetration? (GATE-PI-02)

(a) 5.34 mm (b) 2.15 mm
(c) 4.23 mm (d) 3.85 mm

18. A 900 mm long steel plate is welded by manual metal arc welding process using welding current of 150 A, arc voltage of 20 V and welding speed of 300 mm/min. If the process efficiency of is 0.8 and surface resistance is 36 micro-ohm, the heat input will be

(a) 600 J/mm (b) 480 J/min
(c) 146 kJ (d) 116 kJ

19. The cold cracking susceptibility of the heat affected zone in an arc weld is influenced by (GATE-PI-03)

P. Entrapped hydrogen
Q. Residual stresses
R. Martensitic transformation
S. Slag inclusion

(a) P, S (b) P, Q, S (c) P, Q, R (d) R, S

20. Which of the following process can be used for welding of Aluminum alloys? (GATE-PI-03)

P. Submerged arc welding
Q. Gas metal arc welding
R. Electroslag welding
S. Gas tungsten arc welding

(a) P, Q (b) Q, S (c) Q, R (d) R, S

21. Match the following:

(GATE-PI-03)

Group-I

- P. Arc welding
Q. Friction welding
R. Solid state welding
S. Laser welding

Group-II

1. Diffusion
2. Polarity
3. Focusing
4. Kinetic energy

	P	Q	R	S
(a)	4	1	3	2
(b)	3	2	4	3
(c)	1	2	4	3
(d)	2	4	1	3

22. Match List-I (welding defects) with List-II (causes) and select the correct answer using the codes given below the lists (GATE-PI-03)

List - I

- A. Spatter
B. Distortion
C. Slag inclusion

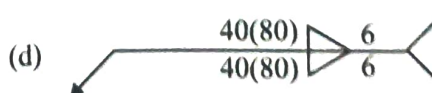
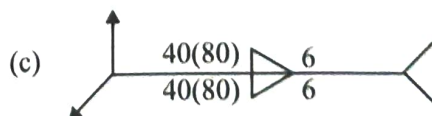
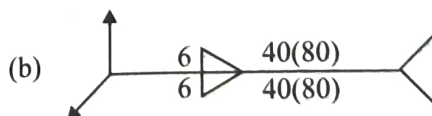
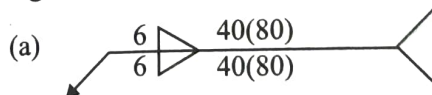
List - II

1. Damp electrodes
2. Arc blow
3. Improper cleaning in multi pass welding
4. Poor joint selection

- D. Porosity
(a) A-4, B-2, C-3, D-1
(b) A-4, B-2, C-1, D-3
(c) A-2, B-4, C-1, D-3
(d) A-2, B-4, C-3, D-1

23. In a fabrication shop, a T-joint is to be made between two plates using a chain intermittent, double fillet weld of 6 mm leg length. The intermittent welds are 40 mm long and spaced 80 mm apart, edge to edge. The welding symbol of the joint is

(GATE-PI-05)



24. Two 8 mm thick steel plates are placed 5 mm apart and welded by a butt joint. Welding is carried out at 20 V and speed of 5 mm/sec. Heat transfer efficiency is 0.80. If the heat required to melt steel is 10 J/mm^3 and melting efficiency is 0.625. The weld current (in Amperes) will be (GATE-PI-05)
- (a) 100 (b) 200 (c) 300 (d) 400

25. Match List – I (Welding problems) with List – II (Causes) and select the correct answer using the codes given below the lists: (GATE-PI-05)

List – I (Welding problems)

- A. Cracking of weld metal
B. Cracking of base metal
C. Porosity
D. Inclusions

List – II (Causes)

1. Excessive stresses
2. High joint rigidity
3. Failure to remove slag from previous deposit
4. Oxidation
5. Excessive $\text{H}_2, \text{O}_2, \text{N}_2$ in the welding atmosphere

Codes:	A	B	C	D
(a)	2	1	5	3
(b)	3	4	2	1
(c)	2	4	5	3
(d)	3	1	4	2

26. Consider the following statements: (GATE-PI-05)
The size of the heat affected zone (HAZ) will increase with

1. Increased starting temperature
2. Increased welding speed
3. Increased thermal conductivity of the base metal
4. Increase in base metal thickness

Which of the statements given above are correct ?

- (a) 1, 2 and 3 (b) 1 and 3
(c) 1 and 4 (d) 2 and 3

27. Consider the following statements: (GATE-PI-05)
The magnitude of residual stresses in welding depends upon

1. Metal melted / deposited
2. Design of weldment
3. Support and clamping of components
4. Welding process used

Which of the statements given above are correct ?

- (a) 1, 2 & 4 (b) 1, 2 & 3 (c) 1 & 3 (d) 2 & 3

28. **Assertion(A):** A sound welded joint should not only be strong enough but should also exhibit a good amount of ductility.

Reason(R): Welding process is used for fabricating mild steel components only. (GATE-PI-05)

29. **Assertion(A):** In electron beam welding process, vacuum is an essential process parameter.

Reason (R): Vacuum provides a highly efficient shield on weld zone. (GATE-PI-05)

30. The DC power source for arc welding has the characteristic $3V + I = 240$, where V = Voltage and I = current in amp. For maximum arc power at the electrode, voltage should be set at (GATE-PI-07)
- (a) 20 V (b) 40 V (c) 60 V (d) 80 V

31. Two pipes of inner diameter 100 mm and outer diameter 110 mm each are joined by flash butt welding using 30 V power supply. At the interface, 1 mm of material melts from each pipe which has a resistance of 42 Ohms. If the unit melt energy is 64.4 MJ/m^3 , then time required for welding (in S) is (GATE-PI-08)

- (a) 1 (b) 5 (c) 10 (d) 20

32. Aluminum strips of 2 mm thickness are joined together by resistance spot welding process by applying an electric current 6000 A for 0.15 sec. The heat required for melting aluminum is 2.9 J/mm^3 . The diameter and thickness of the weld nugget are found to be 5 mm and 2.5 mm respectively. Assuming the electrical resistance to be 75 micro-ohms, the percentage of total energy utilized in forming the weld nugget is (GATE-PI-08)
- (a) 28 (b) 35 (c) 65 (d) 72

32. Which pair among the following solid state welding processes uses heat from an external source?
(GATE-PI-08)

- P- Diffusion welding
Q- Friction welding
R- Ultrasonic welding
S- Forge welding

(a) P & R (b) R & S (c) Q & S (d) P & S

33. Which of the following powders should be fed for effective oxy-fuel cutting of stainless steel?
(GATE-PI-08)

(a) Steel (b) Aluminum (c) Copper (d) Ceramic

34. Autogeneous gas tungsten arc welding of a steel plate is carried out with welding current of 500 A, voltage of 20 V, and weld speed of 20 mm/min. Consider the heat transfer efficiency from the arc to the weld pool as 90%. The heat input per unit length (in kJ/mm) is
(GATE-PI-09)

(a) 27 (b) 35 (c) 45 (d) 55

35. During a steady gas metal arc welding with direct current electrode positive polarity, the welding current, voltage and weld speed are 150 A, 30 V and 6 m/min respectively. A metallic wire electrode of diameter 1.2 mm is being fed at a constant rate of 12 m/min. The density, specific heat and melting temperature of the wire electrode are 7000 kg/m³, 500 J/kg and 1530°C respectively. Assume the ambient temperature to be 30°C and neglect the latent heat of melting. Further consider that 2/3rd of the total electrical power is available for melting of the wire electrode. The melting efficiency (in percentage) of the wire electrode is (GATE-PI-10)

(a) 39.58 (b) 45.25 (c) 49.38 (d) 54.98

36. In a DC arc welding operating, the length characteristic was obtained as $V_{arc} = 20 + 5l$ where the arc length l was varied between 5 mm and 7 mm. Here V_{arc} denotes the arc voltage in Volts. The arc current was varied from 400 A to 500 A. Assuming

linear power source characteristic, the open circuit voltage and the short circuit current for the welding operation are
(GATE-ME & PI-12)

(a) 45 V, 450 A (b) 75 V, 750 A
(c) 95 V, 950 A (d) 150 V, 1500 A

38. For spot welding of two steel sheets (base metal) each of 3 mm thickness, welding current of 10000 A is applied for 0.2 s. The heat dissipated to the base metal is 1000 J. Assuming that the heat required for melting 1 mm³ volume of steel is 20 J and inter-facial contact resistance between sheets is 0.0002 Ω, the volume (in mm³) of weld nugget is ____ (GATE-ME-14-SET-4)

39. A butt weld joint is developed on steel plates having yield and ultimate tensile strength of 500 MPa and 700 MPa, respectively. The thickness of the plates is 8 mm and width is 20 mm. Improper selection of welding parameters caused an undercut of 3 mm depth along the weld. The maximum transverse tensile load (in kN) carrying capacity of the developed weld joint is ____ (GATE-ME-14-SET-3)

40. In an arc welding operation carried out with a power source maintained at 40 volts and 400 Amperes, the consumable electrode melts and just fills the gap between the metal plates to be butt-welded. The heat transfer efficiency for the process is 0.8, melting efficiency is 0.3 and the heat required to melt the electrode is 20 J/mm³. If the travel speed of the electrode is 4 mm/s, the cross-sectional area, in mm², of the weld joint is ____ (GATE-PI-14)

41. A DC welding power source has a linear voltage-current (V-I) characteristic with open circuit voltage of 80 V and a short circuit current of 300 A. For maximum arc power, the current (in Amperes) should be set as ____ (GATE-15-Set 1)

42. During a TIG welding process, the arc current and arc voltage were 50 A and 60 V, respectively, when the welding speed was 150 mm/min. In another

process, the TIG welding is carried out at a welding speed of 120 mm/min at the same arc welding voltage and heat input to the material so that weld quality remains the same. The welding current (in A) for this process is (GATE -15 -Set 2)

- (a) 40.00 (b) 44.72 (c) 55.90 (d) 62.25

43. Two aluminum alloy plates each 10 mm thick and 1 m long are welded without crowning by multi-pass tungsten inert gas butt welding. The joint configuration is V-type with 60° angle and root gap is maintained at 5 mm. If electrode of 5 mm diameter with 500 mm length is used for welding, then the number of electrodes required is

(GATE - PI -15)

- (a) 7 (b) 9 (c) 11 (d) 13

44. The voltage-length characteristic of a direct current arc in an arc welding process is $V = (100 + 40l)$, where l is the length of the arc in mm and V is arc voltage in volts. During a welding operation, the arc length varies between 1 and 2 mm and the welding current is in the range 200-250 A. Assuming a linear power source, the short circuit current is ____ A.

(GATE - 16 - SET - 2)

45. Spot welding of two steel sheets each 2 mm thick carried out successfully by passing 4 kA of current for 0.2 seconds through the electrodes. The resulting weld nugget formed between the sheets is 5 mm in diameter. Assuming cylindrical shape for the nugget, the thickness of the nugget is ____ mm.

Latent heat of fusion for steel	1400 kJ/kg
Effective resistance of the weld joint	200 $\mu\Omega$
Density of steel	8000 kg/m ³

(GATE - 16 - SET - 3)

46. In linear gas tungsten arc welding of two plates of the same material, the peak temperature T (in K) is given as $T = \frac{c_1 q}{\alpha}$, where q is the heat input per unit

length (in J/m) of weld, α is the thermal diffusivity (in m²/s) of the plate materials and C_1 is a constant independent of process parameters and material types.

Two welding cases are given below.

Case I: $V = 15$ V, $I = 200$ A, $v = 5$ mm/s,
 $k = 150$ W/mK, $\rho = 3000$ kg/m³,
 $C = 900$ J/kg-K

Case II: $V = 15$ V, $I = 300$ A, $v = 10$ mm/s,
 $k = 50$ W/mK, $\rho = 8000$ kg/m³,
 $C = 450$ J/kg-K

Where, V is welding voltage, I is welding current, v is welding speed, and k , ρ and C refer to the thermal conductivity, the density and the specific heat of the plate materials, respectively. Consider that electrical energy is completely converted to thermal energy. All other conditions remain same.

The ratio of the peak temperature in Case I to that in Case II is (GATE - PI-16)

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) 1 (d) 2

47. Two flat steel sheets, each of 2.5 mm thickness, are being resistance spot welded using a current of 6000 A and weld time of 0.2 s. The contact resistance at the interface between the two sheets is 200 $\mu\Omega$ and the specific energy to melt steel is 10×10^9 J/m³. A spherical melt pool of diameter 4 mm is formed at the interface due to the current flow. Consider that electrical energy is completely converted to thermal energy. The ratio of the heat used for melting to the total resistive heat generated is ____ (GATE - PI-16)

48. A welding operation is being performed with voltage = 30 V and current = 100 A. The cross-sectional area of the weld bead is 20 mm². The work-piece and filler are of titanium for which the specific energy of melting is 14 J/mm³. Assuming a thermal efficiency of the welding process 70 %, the welding speed (in mm/s) is ____ (correct to two decimal places). (GATE - 18 - SET - 2)

Two metallic sheets are spot welded by passing a current of 8000 A for 0.2 s. Assume that a cylindrical nugget of 8 mm diameter and 3 mm depth is formed. The density of the nugget is 7500 kg/m^3 , effective resistance of the total system is 222 micro-Ohm and heat required to produce 1.0 gram of nugget is 1400 J. The percentage of heat actually utilized in producing the nugget is _____. (PI_GATE - 18)

A gas tungsten arc welding operation is performed using a current of 250 A and an arc voltage of 20 V at a welding speed of 5 mm/s. Assuming that the efficiency is 70%, the net heat input per unit length of the weld will be _____ kJ/mm (round off to one decimal place). (GATE - 19 - SET - 2)

The heat transfer efficiency in arc welding of a plate using a current of 250 A at 20 V is 90%. The heat required to melt the material is 10 J/mm^3 . If the cross-sectional area of the weld joint is 30 mm^2 and the travel speed is 5 mm/s, the melting efficiency (in %) is _____ (round off to 2 decimal places). (GATE - 19_PI)

The heat generated in a resistance spot welding operation for joining two metal sheets with a certain set of process parameters is 2000 J. For a second spot welding operation on the same sheets without any change in the overall resistance of the system, the current is increased by 25% and the time for which the current is applied is reduced to half. The heat generated in the second operation (rounded off to one decimal place) in J is _____. (GATE-PI-20)

The resistance spot welding of two 1.55 mm thick metal sheets is performed using welding current of 10000 A for 0.25 s. The contact resistance at the interface of the metal sheets is 0.0001Ω . The volume of weld nugget formed after welding is 70 mm^3 . Considering the heat required to melt unit volume of metal is 12 J/mm^3 , the thermal efficiency of the welding process is _____ % (round off to one decimal place). (GATE-21_SET-1)

54. A spot welding operation performed on two pieces of steel yielded a nugget with a diameter of 5 mm and a thickness of 1 mm. The welding time was 0.1 s. The melting energy for the steel is 20 J/mm^3 . Assuming the heat conversion efficiency as 10%, the power required for performing the spot welding operation is _____ kW (round off to two decimal places). (GATE-21_SET-2)

55. In an arc welding process, the DC power source characteristic is linear with an open circuit voltage of 60 V and short circuit current of 600 A. The heat required for melting a metal during the welding is 10 J/mm^3 , and the heat transfer and melting efficiencies are 80% and 25%, respectively. If the weld cross-sectional area of 20 mm^2 is made using the maximum arc power, then the required welding speed in mm/s is _____ [round off to one decimal place] (GATE_PI-21)

56. Two mild steel plates of similar thickness, in butt-joint configuration, are welded by gas tungsten arc welding process using the following welding parameters.

Welding voltage	20 V
Welding current	150 A
Welding speed	5 mm/s

A filler wire of the same mild steel material having 3 mm diameter is used in this welding process. The filler wire feed rate is selected such that the final weld bead is composed of 60% volume of filler and 40% volume of plate material. The heat required to melt the mild steel material is 10 J/mm^3 . The heat transfer factor is 0.7 and melting factor is 0.6. The feed rate of the filler wire is _____ mm/s (round off to one decimal place).

(GATE-22_SET-1)

57. In a direct current arc welding process, the power source has an open circuit voltage of 100 V and short circuit current of 1000 A. Assume a linear relationship between voltage and current. The arc voltage (V) varies with the arc length (l) as $V = 10 + 5l$, where V is in volts and l is in mm. The maximum available arc power during the process is _____ kVA (in integer). (GATE-22_SET-2)

58. In a gas tungsten arc welding process under steady state condition, the input voltage and current are measured as 18 V and 160 A, respectively. Heat loss during creation of arc is 40% of the input power. Heat loss through convection and radiation from the workpiece is 800 W. The effective power (in W) utilized to melt the workpiece is _____. [round off to nearest integer] (GATE_PI-22)

59. An arc welding operation is performed at 25 V and 200 A at welding speed of 2 mm/s. The heat used for melting is 80 % of the total heat generated. The unit melting energy of the metal to be joined is 10 J/mm³. The volume of the weld metal produced per unit time, in mm³/s, is _____ (in integer). (GATE_PI-23)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (b)	02. (d)	03. (b)	04. (d)	05. (*)
06. (b)	07. (d)	08. (*)	09. 2.5	10. (a)
11. (a)	12. (a)	13. (b)	14. (b)	15. (d)
16. (c)	17. (c)	18. (a)	19. (b)	20. (a)
21. (*)	22. (b)	23. (d)	24. (*)	25. (*)
26. (d)	27. (*)	28. (*)	29. (*)	30. (c)
31. (c)	32. (*)	33. True	34. True	35. (a)
36. (d)	37. (b)	38. (d)	39. (a)	40. (c)
41. (d)	42. (d)	43. (c)	44. (a)	45. (d)
46. (d)	47. (d)	48. (d)	49. (b)	50. (d)

51. (c)	52. (c)	53. (b,c)	54. 810	
TWO MARK QUESTIONS				
01. (*)	02. (*)	03. (a)	04. (d)	05. (d)
06. (c)	07. (b)	08. (a)	09. (c)	10. (b)
11. (a)	12. (a)	13. (*)	14. (a)	15. (*)
16. (c)	17. (d)	18. (d)	19. (c)	20. (b)
21. (d)	22. (d)	23. (d)	24. (b)	25. (a)
26. (b)	27. (b)	28. (c)	29. (b)	30. (b)
31. (c)	32. (b)	33. (d)	34. (a)	35. (a)
36. (a)	37. (c)	38. 150	39. 70	40. 48
41. 149 to 151		42. (a)	43. (c)	44. 425
45. 2.91	46. (a)	47. (0.2327)		48. 7.5
49. 55.67 %		50. 0.7	51. 33	
52. 1562.5		53. 33.6	54. 39.26	55. 9
56. 10.69		57. 25	58. 928	59. 400

One Mark Solutions

01. Ans: (b)

Sol: By increasing welding current the heat generated will increase and by reducing welding speed more concentration is taking place, hence depth of penetration of the weld will increase.

02. Ans: (d)

Sol: With small variation in arc length the Manual arc welding provides nearly constant current. In manual arc welding welder cannot weld if the melting rate changes drastically with the arc gap. Constant current characteristics give almost constant melting rate.

03. Ans: (b)

Sol: The amount of oxygen consumed from the oxygen cylinder in neutral flame is 1:1 ratio, when the acetylene consumption is 10 liters then oxygen consumption from the oxygen cylinder is also 10 liters.

14. Ans: (d)

Sol: The required current for joining of 1.5 mm thick steel plates is 10,000 Amps.

15. Ans: Maximum & flyer plate

16. Ans: (b)

Sol: In DCSP the work is positive and electrode is negative. As work is positive the heat generated at the work is high, depth of penetration is high. Hence high melting point and high thickness plates can be easily welded but due to lower heat generation at electrode the melting rate of electrode is low and so deposition rate is low.

17. Ans: (d)

Sol: Coatings are given on the electrodes to remove oxides, stabilize the arc, absorb gasses present in the weld pool, producing shielding gasses, provide alloying elements etc.

18. Ans: 1:1

Sol: For the neutral flame the amount of oxygen consumed from the oxygen cylinder is equal to the acetylene consumed hence the ratio of amount oxygen to amount of acetylene is 1:1.

19. Ans: 2.5

Sol: In the case of thin walled, hollow cylindrical parts, maximum length permissible is 2 to 2.5 times the part diameter.

Note: In the case of higher wall thickness, the maximum length can be 4 (or) more times the part diameter.

20. Ans: (a)

Sol: Preheating prevents the hardening of weld bead.

21. Ans: (a)

Sol: When the plate thickness is changed, the amount of heat generated has to be changed and for this to happen the current has to be adjusted.

12. Ans: (a)

Sol: This is due to short supply of oxygen.

13. Ans: (b)

Sol: GTAW is also called as TIG welding which is a non-consumable electrode welding.

14. Ans: (b)

Sol: At the intersection of inner and outer cone the maximum temperature induced is

- (i) 3260°C in neutral flame
- (ii) 3380°C in oxidizing flame and
- (iii) 3040°C in carburizing flame

15. Ans: (d)

Sol: With increase of joint gap the strength of the brazed joint first increases due to increase in wetting area and reduces afterwards due to lower strength of the filler rod material.

16. Ans: (c)

Sol: Friction welding is solid state welding process in which joining is made by conversion of frictional energy into heat and simultaneous application of axial force.

Note:

1. The various solid state welding processes are as follows: friction welding, explosive welding, ultrasonic welding.
2. Solid State Welding is a welding process, in which two work pieces are joined under a pressure providing an intimate contact between them and at a temperature essentially below the melting point of the parent material. Bonding of the materials is a result of diffusion of their interface atoms.

17. Ans: (c)

Sol: GTAW is also called as TIG welding which is a non-consumable electrode welding.

18. Ans: (a)

Sol: Aluminum and magnesium are the highly reactive metals hence it is difficult to weld these metals.

19. Ans: (b)

Sol: In solid state welding like cold pressure welding operation, when large amount of force is applied on the metals the plastic deformations of metals will take place and if the plates are moved slightly, the contaminated layer will be removed and the surfaces become cleaned so that it produces good bond.

20. Ans: (a)

Sol: HAZ means, it is the zone where the parent metal will not melt but the temperature of the metal is above upper critical temperature. Hence the metal will undergo micro structural changes.

21. Ans: Oxidizing flame

Sol: Because during welding with neutral or carburizing flame the zinc present in the brass will get evaporated. But with oxidizing flame, during welding the excess oxygen present in the flame will combine with the some quantity of zinc and form zinc-oxide which floats on the weld pool and does not allow remaining zinc to evaporate.

22. Ans: (b)

Sol: Metal can be transferred by three methods:

1. **Spray transfer:** Small, molten metal droplets from the electrode are transferred to the weld area, at a rate of several hundred droplets per second; the transfer is spatter free and very stable. High DC currents and voltages and large-diameter electrodes are used, with argon or an argon-rich gas mixture

as the shielding gas. The average current required in this process can be reduced by using a pulsed arc, superimposing high-amplitude pulses onto a low, steady current. The process can be used in all welding positions.

2. **Globular transfer:** Carbon-dioxide-rich gases are utilized, and globules are propelled by the forces of the electric-arc transfer of the metal, resulting in considerable spatter. High welding currents are used, making it possible for deeper weld penetration, and welding speeds are higher than in spray transfer. Heavier sections are commonly joined by this method.

3. **Short circuiting:** The metal is transferred in individual droplets (more than 50/s) as the electrode tip touches the molten weld metal and short circuits. *Low currents and voltages are utilized, with carbon-dioxide-rich gases, and electrodes made of small-diameter wire; the power required is about 2 kW.*

23. Ans: (d)

Sol: With increase of joint gap the strength of the brazed joint first increases due to increase in wetting area and reduces afterwards due to lower strength of the filler rod material.

24. Ans: A-3, B-4, C-1, D-2

Sol: In Thermit welding the heat required for melting of the plates is obtained by using exothermic chemical reaction, in projection welding (it is similar to the spot welding used for joining of projected component for flat component) the heat required is generated by using electrical resistance, in MIG welding the heat required is generated due to electric arc and in friction welding the heat is generated due mechanical work.

25. Ans: A. Non-consumable, B. Bare

26. Ans: (d)

Sol: Preheating is the application of heat to the metal before it is welded. This process helps to reduce cracking, hardness, distortion and stresses.

27. Ans: (*) Higher ; Lower

Sol: Welding current should be higher and Weld time should be less. Spot welding is a resistance welding and in resistance welding because of very high temperature induced if time for welding is high, the heat transfer will take place to the thickness direction of the plates which may promote the melting of total thickness of plates which is not preferable and for inducing required amount of heat it is required to supply large quantity of current.

28. Thermit welding

Sol: In Thermit welding, the heat required for melting of plates is obtained due to exothermic chemical reaction and this method is used in remote areas like hill areas and forest areas for joining of rails.

29. Brazing or braze welding

Sol: Brazing or Braze welding is a operation of joining two similar or dissimilar metals without melting of the parent metal but with supply of molten metal of filler rod.

30. Ans: (c)

Sol: For adhesive bonding the area of contact between the two plates must be very high, hence lap joint will give the highest joining area.

31. Ans: (c)

Sol: Rutile coatings are given on the electrodes for producing slag over the weld bead. Low carbon steels are prone to form oxides, to eliminate it requires the slag on the weld bead.

32. Ans: A. Not used, B. Square of the current

Sol: Heat generated (Q) = $I^2 R t$
 $\Rightarrow Q \propto I^2$

33. True

Sol: Because steels have higher melting point, if oxidizing flame is used it is possible to melt and the carbon present in the steel will act as deoxidizer.

34. True

Sol: Refer to the solution of Q.No. 33.

35. Ans: (a)

Sol: The fed *steel powder* will be burned for producing heat required to raise the temperature of plates.

36. Ans: (d)

Sol: Electrons generated at the electron gun are collected and focused by using *magnetic lens*.

37. Ans: (b)

Sol: In resistance welding it is required that R_1 , R_2 , R_3 must be as low as possible and R_4 must be as high as possible. This R_4 is coming from the contact resistance of the joining surfaces or also called as asperities between the joining surfaces.

38. Ans: (d)

Sol: The current required for joining of 1.5 mm thick steel plates is 10000 Amps, for 1 mm thick steel plates the current is less than 10000 Amps and for 3 mm thick steel plate the current is more than 10000 Amps.

39. Ans: (a)

Sol: In thermit welding, *Iron oxide + aluminum powder* will chemically react and produce aluminum oxide which is used as slag, iron which is used as filler material and generate heat which is used as heat for melting of plates.

40. Ans: (c)

Sol: The size of the LASER is very small, hence HAZ also small.

41. Ans: (d)

Sol: Because the electrode has to continuously travel, hence it should be in the circular disc form.

42. Ans: (d)

Sol: Brazing and soldering are the non-fusion welding operations i.e., parent metal will not be melted, hence the joint formation is heterogeneous.

43. Ans: (c)

Sol: Heat input, $Q = CVI$

where,

C = fraction of time during which arc is on

$$= \frac{L}{v} = \frac{\text{length of weld}}{\text{welding speed}}$$

$$\frac{Q}{L} = \frac{VI}{v}$$

$$\Rightarrow \frac{Q}{L} \propto \frac{1}{v}$$

44. Ans: (a)

Sol: Autogenous welding is one in which the metals are joined *without use of filler material*.

45. Ans: (d)

Sol: In Fusion welding for melting the parent material the temperature should be greater than the melting point of the metal hence it is high. Whereas brazing and soldering are the non-fusion welding operations hence the temperature should be less than the melting point of the metal. Brazing temp is above 427°C and soldering is below 427°C .

46. Ans: (d)

Sol: $V_2 = 2V_1$

$$A_{w1} \times V_1 = A_{w2} \times V_2$$

$$A_{w2} = \frac{A_{w1} \times V_1}{V_2} = \frac{A_{w1} \times V_1}{2V_1} = \frac{A_{w1}}{2}$$

$$\% \text{ change} = \frac{A_{w2} - A_{w1}}{A_{w1}} = \frac{0.5 - 1}{1} = -0.5$$

$$= -50\% \rightarrow \text{reduced by } 50\%$$

47. Ans: (d)

Sol: Pure tungsten electrodes are frequently coated with oxides of Th, Zr, La, and Ce. These oxides are expected to perform two important functions

- (a) increase arc stability and
- (b) increase the current carrying capacity of the electrodes.

48. Ans: (d)

Sol: Ratio of oxygen to acetylene

= 1 for neutral flame

= 1.15 to 1.5 for oxidizing flame

= 0.85 to 0.95 for carburising or reducing flame.

49. Ans: (b)

Sol: Heat flux means power input in terms of heat per unit area. It will be maximum in Laser beam welding.

50. Ans: (d)

Sol: In friction welding melting of base materials is not involved.

51. Ans: (c)

Sol: Higher the intensity of heat source, lower the weld pool size, lower the welding temperature level is HAZ. Laser beam and electron beam are very high intensity sources. TIG & MIG have higher intensity and lower weld pool size than SAW.

52. Ans: (c)

Sol: Earing is one of the major defects observed during deep drawing process due to anisotropic nature of sheet metal.

53. Ans: (b, c)

Sol: Brazing: It is non-fusion welding process where the two surfaces are not melting to create a joint.

Adhesive bonding: Metals and non-metals can be joined by this method. A common example of adhesive bonding is plywood, where several layers of wood are bonded with glue.

Arc welding and spot welding are fusion welding process where joint is produced by melting of two surfaces.

54. Ans: 810

Sol: $I = 4500 \text{ A}$,

$T = 0.2 \text{ sec}$

$R = 400 \times 10^{-6} \Omega$

$\eta = 50\%$

Heat required (Joule) = ?

Efficiency, $\eta = \frac{HR}{HG}$

$$0.50 = \frac{HR}{I^2 R \tau}$$

$$0.50 = \frac{H.R}{4500^2 \times 400 \times 10^{-6} \times 0.2}$$

Heat required = 810 J

Two Marks Solutions

01. Ans: A-2, B-4, C-6, D-5

Sol:

- Thermit welding is a process that employs an exothermic reaction to develop a high temperature.
- In resistance seam welding heat is generated due to resistance. ($H = I^2 R t$)
- In MIG welding electric arc is used as heat source.
- Friction welding is a solid state welding process that generates heat through mechanical friction between workpieces in relative to another, with the addition of a lateral force to plastically displace and fuse the materials.

02. Ans: A-3, B-1, C-4, D-2

Sol:

- TIG welding is specially used for welding light alloys and nonferrous materials like aluminium, copper and magnesium.
- Submerged Arc Welding is mainly used to join thick plates of low carbon and low alloy steels and finds applications in ships, pressure vessels, bridges, structural members and nuclear reactors.
- Friction welding is used for welding of drill bits to shanks, engine valve heads to stem, automobile rear axle hub end to axle casing.
- Electron beam welding has intrinsic capability for making deep penetration (heavy sections) welds in a single pass because the electron beam is narrow and its energy density is very high.

03. Ans: (a)

Sol: Heat generated = $I^2 R \tau$

$$= 15000^2 \times 0.0001 \times 0.25$$

$$= 5625 \text{ (in W-s = J)}$$

04. Ans: (d)

Sol:

- Aluminum is welded by TIG or also called GTAW.
- Atomic hydrogen welding issued for welding plain carbon steels, stainless, aluminium, copper and nickel. It is also used for hard facing (or) surfacing of dies and tools and welding of heat resisting alloys.
- An electric soldering gun works well for soldering copper wires. These special soldering tools provide the correct amount of heat for electrical applications.
- The application of laser techniques to a metal such as titanium alloys, which requires extreme cleanliness for attainment of sound welds, is of great interest to the aerospace and chemical industries.

05. Ans: (d)

Sol: Heat generated = $I^2 R \tau$
 $= 5000^2 \times 200 \times 10^{-6} \times 0.2$
 $= 1000 \text{ (in J = W-s)}$

06. Ans: (c)

Sol: Heat generated = Heat utilized

$$I^2 R \tau = m \times h_{fg}$$

$$I^2 R \tau = \rho \times \text{volume} \times h_{fg}$$

$$I^2 \times 200 \times 10^{-6} \times 0.1$$

$$= 8000 \times \frac{\pi}{4} (5)^2 \times 1.5 \times 10^{-9} \times 1400 \times 10^3$$

$$\Rightarrow I = 4061.2 \text{ Amp}$$

07. Ans: (b)

Sol: Heat input = $\frac{VI \text{ (in J/sec = W)}}{\text{speed of welding (mm/sec)}} \times \eta$
 $= \frac{25 \times 300}{8} \times 0.85 = 796.8$
 $= 797 \text{ J/mm}$

08. Ans: (a)

Sol: $V = a + bL$,
 $V_{\text{drop}} = V_o - \frac{V_o}{I_s} \times I_m$

Where V_o = open circuit voltage,
 I_s = short circuit current,
 I_m = measured current,
 V = linear voltage,

At equilibrium, $V = V_{\text{drop}}$

$$a + bL = V_o - \frac{V_o}{I_s} \times I_m$$

At $L = 5 \text{ mm}$, $I_s = 800$,
 $V_o = 80 \text{ V}$, $I_m = 500 \text{ A}$

$$a + b(5) = 80 - \frac{80}{800} \times 500$$

$$a + 5b = 30 \quad \dots\dots\dots(1)$$

At $L = 7 \text{ mm}$, $I_s = 800 \text{ A}$,
 $V_o = 80 \text{ V}$, $I_m = 460 \text{ A}$

$$a + b(7) = 80 - \frac{80}{800} \times 460$$

$$a + 7b = 34 \quad \dots\dots\dots(2)$$

solve (1) & (2)

we get $a = 20$, $b = 2$

$$\therefore V = a + bL = 20 + 2L$$

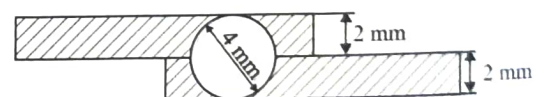
09. Ans: (c)

Sol: Given $I = 10000 \text{ A}$, $\tau = 10 \times 10^{-3} = 0.01 \text{ Sec}$

$$h_{fg} = 300 \text{ kJ/kg}, \quad C_p = 800 \text{ J/kgK}$$

$$\text{Melting point} \quad T_m = 1793 \text{ K}$$

$$\text{Ambient temperature} \quad T_a = 293 \text{ K}$$



Mass of nugget = volume of nugget $\times$ density

$$m = \frac{4}{3} \pi r^3 \times \rho$$

$$m = \frac{4}{3} \pi (0.002)^3 \times 7000$$

$$m = 2.346 \times 10^{-4} \text{ kg}$$

$$\text{Heat required} = m(h_{fg}) + mC_p(T_m - T_a)$$

$$= m(h_{fg} + C_p(T_m - T_a))$$

$$= 2.346 \times 10^{-4} (300 + 0.8(1793 - 293))$$

$$= 351.9 \text{ kJ}$$

$$\begin{aligned}\text{Heat generated} &= I^2 R t \\ &= (10000)^2 \times 500 \times 10^{-6} \times 0.01 \\ &= 500 \text{ kJ}\end{aligned}$$

$$\text{Efficiency } (\eta) = \frac{351.9}{500} = 70.38\%$$

10. Ans: (b)

$$\text{Sol: Net heat input} = \text{Velocity} \times \text{H.R}$$

$$\text{Power} \times \eta_{\text{melting}} \times \eta_{\text{heat transfer}} = \text{Area} \times \text{Velocity} \times \text{H.R}$$

$$2000 \times 0.5 \times 0.7 = 5 \times \text{Velocity} \times 10$$

$$700 = 50 \times \text{Velocity}$$

$$\text{Velocity} = 14 \text{ mm/sec}$$

11. Ans: (a)

$$\text{Sol: Force } (F) = \text{Pressure} \times \text{Area} \times \mu$$

$$= 200 \times \frac{\pi}{4} 10^2 \times 0.5 = 7854 \text{ N}$$

$$\text{Torque} = F \times \frac{3}{4} \times \text{radius}$$

$$\text{Torque} = 7854 \times \frac{3}{4} \times 5 \times 10^{-3} = 29.45 \text{ Nm}$$

$$\text{Power } (P) = \frac{2\pi NT}{60000}$$

$$= \frac{2\pi \times 4000 \times 29.45}{60000} = 12.33 \text{ kW}$$

12. Ans: (a)

Sol: Even though it was not mentioned about the type of flame, but from the given choices the only possible answer is 8 ltrs.

13. Ans: A-4, B-3, C-1, D-2

Cellulose coating: In the arc the coating breaks down to give a voluminous gas shield of H_2 , CO and CO_2 which gives good protection of the molten weld metal with consequent good weld properties.

Rutile coating is one consisting mainly titanium dioxide plus various mineral constituents and small proportion of cellulose. Protection of the metal as it is passed across the arc is provided by the

dissociation of the cellulose and the production of large quantities of slag which will coat the metal droplets as they are transferred.

- Iron powder coating (as alloying agent) may be included in the flux coating to raise the efficiency of the process.
- Basic coating consist of $CaCO_3$ around 43 % and CaF_2 15 to 20 %. These electrodes normally requires baking at temperature of approximately 250°C for 1-2 hrs. Such electrodes produces high quality weld deposits which has high resistance to cracking. This is because hydrogen is removed from weld metal by the action of fluorine.

14. Ans: (a)

Sol: By increasing welding current the heat generated will increase and by reducing welding speed more concentration is taking place hence depth of penetration of the weld will increase.

15. Ans: A-3, B-4, C-2, D-5

Sol: In EBW because of the complete vacuum the weld bead is completely protected from oxidation so highly reactive metal can be welded, SAW is used for joining of plates upto 75 mm thickness in single pass welding, welding of rails in remote areas can be done by using thermit welding and rod can be welded to plates by using friction welding.

16. Ans: (c)

Sol: In oxidizing flame the ratio of oxygen to acetylene is 1.15 to 1.5.

Oxygen consumption for 8 liters of Acetylene is
 $= 1.15 \text{ to } 1.5 \text{ of } 8$
 $= 9.2 \text{ to } 12 \text{ liters.}$

17. Ans: (d)

Sol: Given

$$\rho = 2700 \text{ kg/m}^3$$

$$C_p = 896 \text{ J/kg}$$

$$T_m = 933 - 273 = 660^\circ\text{C}$$

$$T_r = 30^\circ\text{C}$$

Heat supplied = Heat utilized

$$0.5 \text{ J} = m (\text{S.H.} + \text{L.H.})$$

Consider depth of weld be 'h' mm.

$$0.5 = (a \times h) \rho (C_p (T_m - T_r) + h_{fg})$$

$$0.5 = 0.05 \times 10^{-6} \times h \times 2700(896(630) + 398 \times 10^3)$$

$$\Rightarrow h = 0.00385 \text{ m} = 3.85 \text{ mm}$$

18. Ans: (d)

Sol: Welding time = $\frac{900}{300} = 3 \text{ min} = 180 \text{ sec}$

$$I = 150 \text{ A}, \quad V = 20 \text{ V}, \quad \eta = 0.8$$

$$R = 36 \times 10^{-6} \Omega$$

$$\begin{aligned} \text{Heat generated} &= I^2 R \tau \eta \\ &= 150^2 \times 36 \times 10^{-6} \times 180 \times 0.8 \\ &= 116.64 \text{ J} \end{aligned}$$

19. Ans: (c)

Sol: Slag inclusions will never produce cold crack. With entrapped hydrogen the material becomes embrittled which increases the chances of cold cracking, presence of residual stresses will also increase the cold cracking and during welding of steel structures martensite formation takes place which is highly brittle and promotes the cold cracking.

20. Ans: (b)

Sol: In TIG and MIG welding operations the inert gas will be used as shielding gas hence highly reactive metals like aluminium can be easily welded. But sometimes even SAW also may be used.

21. Ans: (d)

Sol: In arc welding by changing the polarity the heat generated at the electrode and work piece can be changed.

22. Ans: (d)

Sol: Spatter is caused due to arc blow, distortion is due to poor joint selection, slag inclusion is due to improper cleaning and porosity is due to damp electrode which induces large quantity of shielding gases.

23. Ans: (d)

Sol:

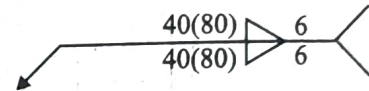


Fig: Standard location of symbols

24. Ans: (b)

Sol: Given, $t = 8 \text{ mm}$, $V = 20 \text{ V}$
 Root gap = 5 mm , Speed = 5 mm/sec
 $\eta_{HT} = 0.8$, $HI = 10 \text{ J/mm}^3$
 $\eta_{melting} = 0.625$, $I = ?$

$$\begin{aligned} \text{Vol. of weld bead} &= t \times \text{root gap} \times L \\ &= 8 \times 5 \times L = 40 \times L \text{ mm}^3 \end{aligned}$$

where, L = length of Weld bead

$$\text{H.R} = 40 L \times 10 = 400 L \text{ (in J)} \dots\dots\dots (1)$$

$$\text{H.I} = \frac{VI \times L}{\text{speed}} = \frac{20 \times I \times L}{5} \text{ (in J)} \dots\dots\dots (2)$$

Equating equations 1 and 2

$$\frac{20 \times IL}{5} \times 0.8 \times 0.625 = 400 L$$

$$\Rightarrow I = 200 \text{ A}$$

25. Ans: (a)

Sol:

- Cracking of weld metal occurs due to high joint rigidity and cracking of base metal occurs due to excessive stresses.
- Porosity is mainly caused due to hydrogen, oxygen, nitrogen.

Inclusions caused due following reasons:

- In multi-pass welding, failure to remove slag from previous deposit, under cut on previous pass, high viscosity of weld metal rapid solidification, insufficient heat input.

26. Ans: (b)

Sol: With increase of welding speed the rate of heat transfer reduces and with increase of thickness of plates also thermal resistance increases, in both of the cases heat transfer will come down hence the heat affected zone (HAZ) will be reduced.

27. Ans: (b)

Sol: The welding process used will never creates the internal residual stresses hence option (b) is correct.

28. Ans: (c)

Sol: The weld bead should exhibit ductility to withstand for impact loads but the welding process is not only used for mild steel but also can be used for other materials.

29. Ans: (b)

Sol: The basic purpose of providing the vacuum in EBM is to avoid the dispersion of electrons and not to provide efficient shield. It can act as shield as additional functionality.

30. Ans: (b)

Sol: Given, $3V + I = 240$

$$I = 240 - 3V$$

$$\text{Power (P)} = VI = V(240 - 3V) \\ = 240V - 3V^2$$

For optimum power

$$\frac{dP}{dV} = 0 \Rightarrow 240 - 6V = 0$$

$$V = \frac{240}{6} = 40 \text{ V}$$

31. Ans: (c)

$$\text{Sol: Volume} = \frac{\pi}{4}(110^2 - 100^2) \times 2 = 3298.66 \text{ mm}^3$$

$$\text{Heat required} = \text{total volume} \times \text{heat rate/mm}^3 \\ = 3298.66 \times 10^{-9} \times 64.4 \times 10^6 \\ = 212.4 \text{ Joules}$$

$$\text{Power (P)} = VI = V \times \frac{V}{R} = \frac{V^2}{R} \\ = \frac{30^2}{42} = 21.43 \text{ J/sec}$$

$$\text{Energy (heat)} = \text{Power} \times \text{time}$$

$$\Rightarrow \text{Time} = \frac{212.4}{21.43} = 10 \text{ sec}$$

32. Ans: (b)

Sol: Given,

$$t = 2 \text{ mm}, \quad I = 6000 \text{ A},$$

$$\tau = 0.15 \text{ sec}, \quad D = 5 \text{ mm},$$

$$\text{Heat rate (H.R)} = 2.9 \text{ J/mm}^3,$$

$$h = 2.5 \text{ mm}, \quad R = 75 \times 10^{-6} \Omega$$

$$\text{Total heat required (HR)} = \text{total volume HR/mm}^3 \\ = \frac{\pi}{4} D^2 h \times 2.9$$

$$= \frac{\pi}{4} 5^2 \times 2.5 \times 2.9 = 142.35 \text{ J}$$

$$\text{Actual heat generated (HG)} = I^2 R \tau$$

$$= 6000^2 \times 75 \times 10^{-6} \times 0.15 \\ = 405 \text{ J}$$

$$\text{Efficiency } \eta = \frac{\text{H.R}}{\text{H.G}} = \frac{142.28}{405} = 35\%$$

33. Ans: (d)

Sol:

- Diffusion welding (DFW) is a solid-state welding process that results from the application of heat and pressure, usually in a controlled atmosphere, with sufficient time allowed for diffusion and coalescence to occur. Temperatures are well below the melting points of the metals (about $0.5 T_m$ is the maximum), and plastic deformation at the surfaces is minimal.
- Forge welding is a welding process in which the components to be joined are heated to hot working temperatures and then forged together by hammer or other means.

34. Ans: (a)

Sol: By feeding the steel powder, the powder will burn for producing heat required for rising the temperature of plates to rise to the kindling temperature.

35. Ans: (a)

Sol: Given, $V = 20 \text{ V}$, $I = 500 \text{ Amp}$,

$$v = \frac{20}{60} \text{ mm/sec}, \eta = 0.9$$

$$\begin{aligned} \text{Power (P)} &= \frac{VI}{\text{speed}} \times \eta \\ &= \frac{20 \times 500}{20/60} \times 0.9 = 27 \text{ kJ/mm} \end{aligned}$$

36. Ans: (a)

Sol: Given, $I = 150 \text{ A}$, $V = 30 \text{ V}$,
 Speed = 6 m/min, Wire dia = 1.2 mm,
 $f = 12 \text{ m/min}$

$$H.G = VI = 150 \times 30 = 4500 \text{ W}$$

$$\begin{aligned} H.R &= \frac{\pi}{4} \times d^2 \times f \times \rho \times C_p \times (T_m - T_r) \\ &= \frac{\pi}{4} (1.2)^2 \times 10^{-6} \times \frac{12}{60} \times 7000 \times 500 (1530 - 30) \\ &= 1187.5 \text{ W} \end{aligned}$$

$$\eta = \frac{H.R}{H.G} = \frac{1187.5}{4500 \times \frac{2}{3}} = 39.58\%$$

37. Ans: (c)

Sol: Given, $V_a = 20 + 5L$, $L = 5 \text{ to } 7 \text{ mm}$,
 $I = 400 \text{ to } 500 \text{ A}$

At equilibrium condition

$$V_a = V_p$$

$$20 + 5L = V_o - \frac{V_o}{I_s} \times I$$

At $L = 5 \text{ mm}$, $I = 500 \text{ A}$

$$20 + 5(5) = V_o - \frac{V_o}{I_s} \times 500$$

$$45 = V_o - \frac{V_o}{I_s} \times 500$$

At $L = 7 \text{ mm}$, $I = 400 \text{ A}$

$$20 + 5(7) = V_o - \frac{V_o}{I_s} \times 400$$

$$55 = V_o - \frac{V_o}{I_s} \times 400 \dots (2)$$

Solve (1) and (2) for V_o and I_s

We get $I_s = 950 \text{ A}$

I_s substitute in equation (1)

$$\Rightarrow V_o = 95 \text{ V}$$

38. Ans: 150

Sol: Given,

$$I = 10000 \text{ A},$$

$$\tau = 0.2 \text{ sec},$$

$$\text{Heat dissipated} = 1000 \text{ J},$$

$$\text{Heat required} = 20 \text{ J/mm}^3,$$

$$\text{Interfacial contact resistance } R = 0.0002 \text{ Ohms}$$

$$\text{Heat generated} = I^2 R \tau = 4000 \text{ J}$$

Heat utilized for welding or melting

$$= 4000 - 1000$$

$$= 3000 \text{ J}$$

$$= 20 \times \text{volume of weld nugget}$$

Volume of weld nugget

$$= 3000/20$$

$$= 150 \text{ mm}^3$$

39. Ans: 70

Sol: Given, $b = 20 \text{ mm}$; $t = 8 \text{ mm}$

Undercut (d) = 3 mm

Due to undercut, the load bearing area = $b \times (t - d)$

$$= 20 \times (8 - 3) = 20 \times 5 = 100 \text{ mm}^2$$

Maximum load carrying capacity of the weld

$$= \sigma_{ut} \times \text{area} = 700 \times 100 = 70 \text{ kN}$$

40. Ans : 48

Sol: Heat generation rate = $V \cdot I$

$$= 40 \times 400 = 16000 \text{ W}$$

Heat utilized

$$= \text{heat generation rate} \times \text{melting Efficiency} \times \text{heat transfer efficiency}$$

$$= 16000 \times 0.8 \times 0.3$$

$$= 3840 \text{ W}$$

Volume filling rate of weld bead

$$= \text{C.S area} \times \text{velocity}$$

$$= (A \times 4) \text{ mm}^3/\text{sec}$$

Rate of heat required

$$= 20 \times 4A = 80 \text{ AWatt}$$

$$= \text{heat utilized}$$

$$A = \frac{3840}{80} = 48 \text{ mm}^2$$

41. Ans: 149 to 151

Sol: Given, $V_o = 80 \text{ V}$, $I_s = 300 \text{ A}$

$$V = V_o - \frac{V_o}{I_s} \times I = 80 - \frac{80}{300} \times I$$

$$P = VI$$

$$= \left(80 - \frac{80}{300} \times I\right) \times I$$

$$P = 80I - \frac{80}{300} I^2$$

For maximum power, $\frac{dP}{dI} = 0$

$$0 = 80 - 2 \times \frac{80}{300} I$$

$$\Rightarrow I = 150 \text{ A}$$

42. Ans: (a)

Sol: Given, $V = 60 \text{ V}$, $I = 50 \text{ A}$

$$\text{Speed} = 150 \text{ mm/min} \rightarrow 1^{\text{st}} \text{ process}$$

$$= 120 \text{ mm/min} \rightarrow 2^{\text{nd}} \text{ process}$$

$$V_1 = V_2$$

$$(H.I)_1 = (H.I)_2$$

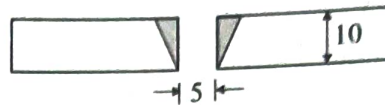
$$\left(\frac{V.I}{\text{Speed}}\right)_1 = \left(\frac{V.I}{\text{Speed}}\right)_2$$

$$I_2 = \frac{I_1}{\text{Speed}_1} \times \text{Speed}_2$$

$$= \frac{50}{150} \times 120 = 40 \text{ A}$$

43. Ans: (c)

Sol:



$$\text{Area of weld bead} = (5 \times 10) + (2 \times 0.5 \times 10 \tan 30^\circ \times 10)$$

$$= 107.73 \text{ mm}^2$$

$$\text{Volume of weld bead} = \text{area} \times \text{length}$$

$$= 107.73 \times 1000$$

$$= 107735 \text{ mm}^3$$

$$\text{Volume per electrode} = \frac{\pi}{4} (5)^2 \times 500 = 9812.5 \text{ mm}^3$$

Number of electrodes required

$$= \frac{107735}{9812} = 10.97 \approx 11 \text{ electrodes}$$

44. Ans: 425 (range 423 to 428)

Sol: Given, $V = 100 + 40 L$, $L = 1 \text{ to } 2 \text{ mm}$,

$$I = 200 \text{ to } 250 \text{ A},$$

At $L = 1 \text{ mm}$, $I = 250 \text{ Amp}$

$$V = 100 + 40 \times 1 = 140 = V_o - \frac{V_o}{I_s} \times 250$$

At $L = 2 \text{ mm}$, $I = 200 \text{ Amp}$

$$V = 100 + 40 \times 2 = 180 = V_o - \frac{V_o}{I_s} \times 200$$

$$\Rightarrow 40 = 50 \times \frac{V_o}{I_s}$$

$$\frac{V_o}{I_s} = \frac{40}{50} = \frac{4}{5}$$

$$V_o = 140 + \frac{4}{5} \times 250$$

$$= 140 + 200 = 340 \text{ V}$$

$$\frac{V_o}{I_s} = \frac{4}{5} \Rightarrow I_s = \frac{V_o \times 5}{4} = \frac{340 \times 5}{4} = 425 \text{ A}$$

45. Ans: 2.91 (range 2.85 to 2.95)

Sol: $I^2 R \tau = \text{volume} \times \rho \times H.R / \text{kg}$

$$4000^2 \times 200 \times 10^{-6} \times 0.2 = \text{volume} \times 8000 \times$$

$$1400 \times 10^3$$

$$\text{Volume} = 5.7 \times 10^{-8} \text{ m}^3 = \frac{\pi}{4} \times 5^2 \times h \times 10^{-9}$$

$$\Rightarrow h = 2.91 \text{ mm}$$

46. Ans: (a)

Sol: Given,

$$\text{Thermal diffusivity } (\alpha) = \frac{\text{heat conducted}}{\text{heat stored}} = \frac{k}{\rho C}$$

$$\alpha_1 = \frac{150}{3000 \times 900} = 55.5 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\alpha_2 = \frac{50}{8000 \times 450} = 13.88 \times 10^{-6} \text{ m}^2/\text{s}$$

$$q_1 = \frac{VI}{v} = \frac{15 \times 200}{5 \times 10^{-3}} = 6 \times 10^5$$

$$q_2 = \frac{15 \times 300}{10 \times 10^{-3}} = 4.5 \times 10^5$$

$$\frac{T_1}{T_2} = \frac{q_1}{q_2} \times \frac{\alpha_2}{\alpha_1} = \frac{6 \times 10^5}{4.5 \times 10^5} \times \frac{13.88 \times 10^{-6}}{55.5 \times 10^{-6}} = \frac{6 \times 13.88}{4.5 \times 55.55} = 0.333 = \frac{1}{3}$$

47. Ans: (0.2327)

Sol: Given

$$t = 2.5 \text{ mm}, \quad I = 6000 \text{ A},$$

$$\tau = 0.2 \text{ sec}, \quad R = 200 \mu\Omega$$

$$\text{Heat required, } (H.R/m^3) = 10 \times 10^9 \text{ J/m}^3$$

$$\text{Nugget diameter } (d) = 4 \text{ mm (sphere)}$$

$$\text{Heat generated} = I^2 R \tau$$

$$= 6000^2 \times 200 \times 10^{-6} \times 0.2 = 1440 \text{ J}$$

$$\text{Heat required} = \text{volume} \times \text{heat required/m}^3$$

$$= \frac{4}{3} \times \pi \times R^3 \times H.R/m^3$$

$$= \frac{4}{3} \times \pi \times (2 \times 10^{-3})^3 \times 10 \times 10^9$$

$$= 335.1032 \text{ J}$$

$$\eta_{H.J} = \frac{\text{Heat required}}{\text{Heat generated}}$$

$$= \frac{335.1032}{1440} = 0.2327$$

48. Ans: 7.5

$$\text{Sol: Given: } V = 30 \text{ V}, \quad I = 100 \text{ A}, \quad A_w = 20 \text{ mm}^2,$$

$$\text{Heat required, } H_m = 14 \text{ J/mm}^3$$

$$\eta_T = 0.7, \quad \text{welding speed, } V_w = ?$$

$$\Rightarrow H_s = \frac{VI}{A_w V_w} \times \eta_T$$

$$H_s = \frac{30 \times 100}{20 \times V_w} \times 0.7$$

$$H_s = \frac{105}{V_w}$$

$$\text{Assuming 100\% melting efficiency, } H_s = H_m$$

$$\Rightarrow \frac{105}{V_w} = 14 \Rightarrow V_w = 7.5 \text{ mm/s}$$

49. Ans: 55.67 %

$$\text{Sol: Given data: } I = 8000 \text{ A}, \quad \tau = 0.2 \text{ sec},$$

$$d = 8 \text{ mm}, \quad h = 3 \text{ mm},$$

$$\rho = 7500 \text{ kg/m}^3, \quad R = 222 \times 10^{-6} \Omega$$

$$\Rightarrow m_n = \rho \times V_n$$

$$= 7500 \times 10^{-9} \times \frac{\pi}{4} \times 64 \times 3 \times 10^3 = 1.13 \text{ gm}$$

$$\text{Heat required for melting,}$$

$$H_m = 1.13 \times 1400 \text{ J} = 1582 \text{ J}$$

$$\text{Heat supplied} = I^2 R \tau = 8000^2 \times 222 \times 10^{-6} \times 0.2 = 2841.6 \text{ J}$$

$$\text{Welding efficiency,}$$

$$\eta = \frac{H_m}{H_s} = \frac{1582}{2841.6} \times 100 = 55.67\%$$

50. Ans: 0.7

$$\text{Sol: Given, GTAW, current } (I) = 250 \text{ A},$$

$$\text{Voltage } (V) = 20 \text{ V},$$

$$\text{Speed } (v) = 5 \text{ mm/sec},$$

$$\text{Arc efficiency } (\eta) = 0.7$$

$$\text{Duty cycle } (D) = 1 \text{ (not given)}$$

$$\text{Heat input per unit length, } \frac{H}{l} = \frac{\eta DVI}{v}$$

$$= \frac{0.7 \times 1 \times 250 \times 20}{5}$$

$$= 700 \text{ J/mm} = 0.7 \text{ kJ/mm}$$

Q. Ans: 33

Sol: Given,
 $\eta_b = 0.9$,
 $V = 20 \text{ V}$,
 $u = 10 \text{ J/mm}^3$,
 $v = 5 \text{ mm/sec}$; $\eta_m = ?$
 $D = 1 \text{ (not given)}$
 $I = 250 \text{ A}$
 $A = 30 \text{ mm}^2$
 $\eta_{DVI} = \frac{v}{uA}$
 $s = \frac{\eta \times 1 \times 20 \times 250}{10 \times 30}$

$$\eta = \frac{5 \times 10 \times 30}{20 \times 250} = 0.3$$

$$\eta_b \times \eta_m = \eta$$

$$0.9 \times \eta_m = 0.3$$

$$\eta_m = \frac{0.3}{0.9} = 0.33 = 33\%$$

Q. Ans: 1562.5

Sol: Given,
 $H.G_1 = 2000 \text{ J}$;
 $I_2 = 1.25 I_1$,
 $\tau_2 = 0.5 \tau_1$
 $H.G_2 = I_2^2 R \tau_2$
 $= (1.25 I_1)^2 R (0.5 \tau_1)$
 $= 1.25^2 \times 0.5 \times I_1^2 R \tau_1$
 $= (1.25)^2 \times 0.5 \times H.G_1$
 $= 1.25^2 \times 0.5 \times 2000 = 1562.5 \text{ Joules}$

Q. Ans: 33.6 (Range 30 to 36)

Sol: Given data,
Resistance spot welding
Sheet thickness (t) = 1.55 mm
Current (I) = 10000 A
Time (τ) = 0.25 s
Contact Resistance (R) = 0.001 Ω
Volume of Nugget (N) = 70 mm³
Heat required to melt unit volume (u)
 $= \frac{H_u}{V} = 12 \text{ J/mm}^3$
Thermal efficiency (η_{th}) = $\frac{\text{heat utilised } (H_u)}{\text{Heat energy supplied } (H_s)}$

$$\text{Heat utilized } (H_u) = u \times v = 12 \times 70 = 840 \text{ J}$$

$$\text{Heat energy supplied } (H_s) = I^2 R \tau$$

$$= (10000)^2 \times 0.0001 \times 0.25$$

$$= 2500 \text{ J}$$

$$\eta_{th} = \frac{H_u}{H_s} = \frac{840}{2500} = 33.6\%$$

54. Ans: 39.26 (Range 39.00 to 40.00)

Sol: Given, Resistance spot welding

$$\text{Nugget Diameter } (D_N) = 5 \text{ mm}$$

$$\text{Nugget thickness } (h_N) = 1 \text{ mm}$$

$$\text{Weld time } (\tau) = 0.15$$

$$\text{Melting energy } (U_m) = 20 \text{ J/mm}^3$$

$$\text{Heat conversion efficiency } \eta_c = 10\% = 0.1$$

$$\text{Required power for spot welding } (P) = ?$$

$$P = I^2 R = ?$$

$$\text{Heat generated } (HG) = I^2 R \tau$$

$$\text{Heat required to melt material } (HR) = V \times U$$

$$= \frac{\pi}{4} \times 5^2 \times 1 \times 20$$

$$\eta_c = \frac{(HR)}{(HG)} = \frac{HR}{P \times \tau}$$

$$P = \frac{HR}{\eta \times \tau} = \frac{\frac{\pi}{4} \times 5^2 \times 1 \times 20}{1.0 \times 0.1} = 39269.90 \text{ J/S}$$

$$P \approx 39.27 \text{ kW}$$

55. Ans: 9 (Range 8.8 to 9.2)

Sol: Given, Open circuit voltage (V_0) = 60 V

$$\text{Short circuit current } (I_s) = 600 \text{ A}$$

$$u = \text{Energy required to melt unit volume} = 10 \text{ J/mm}^3$$

$$\text{Heat transfer } (\eta) = 80\%$$

$$\text{Melting efficiencies } (\eta_m) = 25\%$$

$$\text{Cross-sectional area, } A = 20 \text{ mm}^2$$

$$\text{Arc power at maximum}$$

$$V = V_0 - \frac{V_0}{I_s} \cdot I \Rightarrow V = 60 - \frac{60}{600} I$$

$$\therefore V = 60 - 0.1I$$

$$\text{For Maximum power}$$

$$\frac{dP}{dV} = 0 \Rightarrow \frac{d\{(60 - 0.1I)\}}{dV} = 0$$

$$\text{Power} = VI = (60 - 0.1 I) I = 60 I - 0.1 I^2$$

$$\text{For maximum power } \frac{dP}{dI} = 0$$

$$\Rightarrow 60 - 0.2 I = 0 \Rightarrow I = 300 \text{ A}$$

$$V = 60 - (0.1 \times I) = 60 - (0.1 \times 300) = 30 \text{ V}$$

$$v = \frac{\eta DVI}{uA} \quad \eta = \eta_m \times \eta_h = 0.8 \times 0.25$$

$$D = 1 \text{ (not given)}$$

$$V = 30 \text{ V}, \quad I = 300 \text{ A}$$

$$v = \frac{0.8 \times 0.25 \times 1 \times 30 \times 300}{10 \times 20} = 9 \text{ mm/sec}$$

56. Ans: (10.69)

Sol: Welding current = 150A

Welding voltage = 20V

$$\text{Heat Generated } Q_1 = V \times I = 150 \times 20$$

$$Q_1 = 3000 \text{ Watt}$$

$$\text{Net Heat supplied to the Work Piece } (Q_2) = Q_1 \times \eta_{HT}$$

$$Q_2 = 300 \times 0.7 = 2100 \text{ Watt}$$

$$\text{Heat Required to Melt the material } (Q_3) = Q_2 \times \eta_m$$

$$Q_3 = 2100 \times 0.6 = 1260 \text{ Watt}$$

Since Heat Required to melt the material is given directly in the question = 10^9 J/mm^3

Hence we can write

$$1260 = 10 \times \text{Volumetric metal Deposition Rate (VMDR)}$$

$$\text{VMDR} = \frac{1260}{10} = 126 \text{ mm}^3/\text{s}$$

It means total $126 \text{ mm}^3/\text{s}$ metal need to be deposited in the gap between Work Piece

It is given in the question that 60% of the total volume is to be filled by electrode.

Hence, effective volumetric metal deposition by electrode = $126 \times 0.6 = 75.6 \text{ mm}^3/\text{s}$

Volumetric metal deposition rate = Cross section Area of electrode $\times$ Electrode feed rate

$$75.6 = \frac{\pi}{4} d_e^2 \times \text{feed Rate}$$

$$\text{Feed Rate} = \frac{75.6}{\frac{\pi}{4} (3)^2} = 10.69 \text{ mm/s}$$

57. Ans: 25

Sol: Given, OCV = 100 V

$$\text{SCC} = 1000 \text{ A}$$

$$\text{Voltage Arc length characteristic } V_{\text{arc}} = 10 + 5I$$

We have,

$$V_P = \text{OCV} - \frac{\text{OCV}}{\text{SCC}} I$$

$$V_P = 100 - \frac{100}{1000} I$$

$$V_P = 100 - \frac{I}{10}$$

For stable arc

$$V_{\text{arc}} = V_P$$

$$10 + 5I = 100 - \frac{I}{10}$$

$$I = 900 - 50I$$

$$\text{Arc Power } (P) = V \times I$$

$$P = (10 + 5I)(900 - 50I)$$

$$P = 9000 + 4000I - 250I^2$$

For Maximum Arc Power

$$\frac{dP}{dI} = 0$$

$$\frac{d}{dI} (9000 + 4000I - 250I^2) = 0$$

$$I = 4000/500 = 8 \text{ mm}$$

Hence, for Maximum arc power, arc length is 8 mm

$$\text{Voltage @ } I = 8 \text{ mm} \Rightarrow V = 10 + 5I$$

$$V = 10 + (5 \times 8) = 50 \text{ Volt}$$

Current @ $I = 8 \text{ mm}$

$$I = 900 - 50I$$

$$I = 900 - (50 \times 8)$$

$$I = 900 - 400 = 500 \text{ A}$$

Maximum Arc Power

$$P_{\text{max}} = V \times I$$

$$P_{\text{max}} = 50 \times 500 = 25000 \text{ VA} = 25 \text{ kVA}$$

58. Ans: 928

Sol: Given data:

Voltage = 18 volt

Current = 160 Amp

Welding power or heat generated

$$Q_1 = VI$$

$$Q_1 = 18 \times 160 = 2880 \text{ watt}$$

Since 40% of the input power is lost hence heat supplied to workplace

$$Q_2 = 2880 \times 0.6 = 1728 \text{ Watt}$$

Now, 800 W of heat is lost through the workplace by the mode of convection and radiation, hence effective heat (or power) utilized to melt the workplace

$$Q_3 = Q_2 - Q_{\text{loss}}$$

$$Q_3 = 1728 - 800 = 928 \text{ Watt}$$

59. Ans: 400

Sol: $V = 25 \text{ V},$

$$\eta_m = 80\% = 0.80$$

$$I = 200 \text{ A},$$

$$U = 10 \text{ J/mm}^3$$

$$v = 2 \text{ mm/sec}$$

$$\text{volume/time} = ?$$

η = Heat transfer efficiency η is not given,

$\eta = 100\%$ consider

$$\eta = 1$$

$$\eta = \frac{UAv}{VI}$$

$$0.8 = \frac{10 \times Av}{25 \times 200}$$

$$Av = \text{Volume per unit time} = 400 \text{ mm}^3/\text{sec}$$